Statements in First-Order Logic (FOL) are constructed using following components:

- Predicates: $P(x), Q(x, y), \ldots$
- Logical Connectives: $\sim$, $\wedge, \vee, \Longrightarrow, \Leftrightarrow$
- Quantifiers: The symbols $\forall$ (**Universal**, "for all") and $\exists$ (**Existential**, "there exists").

Definition: An open statement (also called a **predicate**) is a sentence that contains one or more variables and whose truth value depends on the values assigned for the variables. We represent an open statement by a capital letter followed by the variable(s) in parenthesis, e.g., $P(x). Q(x), R(x, y)$, ...etc

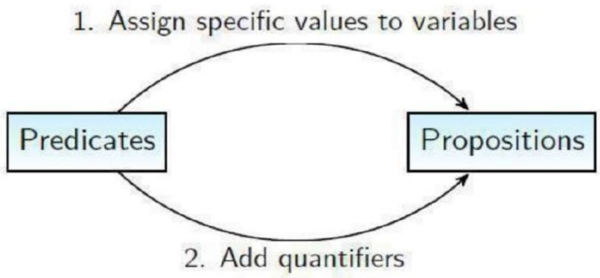
Note:

- A predicate is neither true nor false
- A predicate becomes a proposition when the variables are substituted with specific values
- The domain of a predicate variable is the set of all values that may be substituted for the variable

Definition: The **domain** or **universe** or **universe of discourse** for a predicate variable is the set of values that may be assigned to the variable.

Predicates to propositions

There are two methods to obtain propositions from predicates
1. Assign specific values to variables
2. Add quantifiers



- Quantifiers are words that refer to quantities such as "all" or "some" and they tell for how many elements a given predicate is true
- Two types of quantifiers:
  1. Universal quantifier ($\forall$)
  2. Existential quantifier ($\exists$)

Definition. Let x be a variable whose domain contains the set S, and let $P(x)$ be an open sentence. The *universal quantification* of $P(x)$ over S, written

$$\forall x \in S,\ P(x),$$

is a statement that is true if $P(x)$ is true for each choice of $x \in S$, and it is false if $P(x)$ is false for at least one choice of $x \in S$.

Notation: $\forall x P(x)$, where $\forall$ is called the universal quantifier.

Read as "for all x $P(x)$" or "for every x $P(x)$".

An element x for which $P(x)$ is false is called a counterexample of $\forall x P(x)$.

Other ways to say $\forall$:

- All of...
- For each....
- Given any.....
- For any....
- For arbitrary....

Definition. Let x be a variable whose domain contains the set S, and let $P(x)$ be an open sentence. The *existential quantification* of $P(x)$ over S, written

$$\exists x \in S,\ P(x),$$

is a statement that is true if $P(x)$ is true for at least one choice of $x \in S$, and it is false if $P(x)$ is false for each choice of $x \in S$.

The statement $\exists x \in S,\ P(x)$ reads as "there exists some x in S such that $P(x)$."

Notation: $\exists x P(x)$, where $\exists$ is called the existential quantifier.

Other ways to read $\exists$:

"There is an x such that $P(x)$."

"There is at least one x such that $P(x)$."

"For some x, $P(x)$."

"We can find an x, such that $p(x)$"

1) "For any real number x, $\sqrt{x^2} = |x|$" T

$\forall x \in \mathbb{R}, \sqrt{x^2} = |x|$

$\sqrt{x^2} = |x|$ for all $x \in \mathbb{R}$

For each real number x, $\sqrt{x^2} = |x|$

2) "There exists real number x such that $\sqrt{x^2} = -x$." T

"$\exists x \in \mathbb{R}, \sqrt{x^2} = |x|$"

"$\sqrt{x^2} = |x|$ for some real number"

"There is at least one real number x such that $\sqrt{x^2} = -x$"

3) "There exist ~~real~~ real number x such that $\sqrt{x^2} \neq -x$" T

"$\exists x \in \mathbb{R}, \sqrt{x^2} \neq -x$.

Negations of Quantified Statements (Quantifiers and 'not')

Suppose A is a non-empty set and $P(x)$ where $x \in A$ is a predicate and B is a non-empty subset of A.

Consider the propositions,"not($\forall x \in B, P(x)$)"and "not ($\exists x \in B, P(x)$)."

not($\forall x \in B, P(x)$) means it is not the case that for each and every $x \in B$, we have $P(x)$.

This means, there is some $x \in B$ such that we have not $P(x)$ and this means $\exists x \in B$, not $P(x)$.

$\therefore$ not($\forall x \in B, P(x)$) and $\exists x \in B$, not $P(x)$ are equivalent.

not($\exists x \in B, P(x)$) means it is not the case that there is some $x \in B$ such that we have $P(x)$.

This means, for each and every $x \in B$, we have not $P(x)$, and this means $\forall x \in B$ not $P(x)$.

$\therefore$ not($\exists x \in B, P(x)$) and $\forall x \in B$, not $P(x)$ are equivalent.

Note:

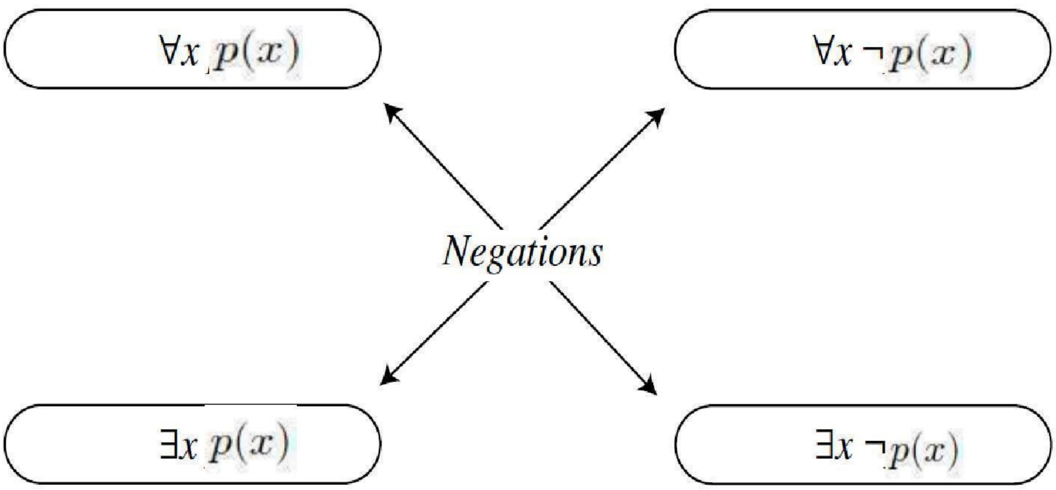
- $\sim (\forall x \in D, p(x)) \equiv \exists x \in D, \sim p(x)$
- $\sim (\exists x \in D, p(x)) \equiv \forall x \in D, \sim p(x)$

Write down the negation of the following statements.

(i) $\sqrt{x^2} = -x$ for all real number x

(ii) $\sqrt{x^2} = -x$ for some real number x

(iii) For each x in $\mathbb{R}^+ \cup \{0\}$, $\sqrt{x^2} = x$



Predicates with conjunction and disjunction

- **Logical Equivalences Involving Quantifiers**

01. $\forall x \in D, P(x) \wedge Q(x)$ and $(\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$ are logically equivalent.

i.e. $\forall x \in D, P(x) \wedge Q(x) \equiv (\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$.

02. $\forall x \in D, P(x) \vee Q(x)$ and $(\forall x \in D, P(x)) \vee (\forall x \in D, Q(x))$ are not logically equivalent.

i.e. $\forall x \in D, P(x) \vee Q(x) \not\equiv (\forall x \in D, P(x)) \vee (\forall x \in D, Q(x))$.

Proof of 01.

First, we show that if $\forall x \in D, P(x) \wedge Q(x)$ is true, then $(\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$ is true. Second, we show that if $(\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$ is true, then $\forall x \in D, P(x) \wedge Q(x)$ is true.

Suppose $\forall x \in D, P(x) \wedge Q(x)$. Let $a \in D$.

Then, $P(a) \wedge Q(a)$ is true. Therefore, $P(a)$ is true and $Q(a)$ is true.

Since $P(a)$ is true for any a in D and $Q(a)$ is true for any a in D, we have $(\forall x \in D, P(x))$ is true and $(\forall x \in D, Q(x))$is true.

Therefore, $(\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$ is true.

Now suppose $(\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$ is true.

Then, for any a in D, both $P(a)$ and $Q(a)$ are true. Hence, $\forall x \in D, P(x) \wedge Q(x)$ is true.

Thus, $\forall x \in D, P(x) \wedge Q(x) \equiv (\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$.

02.. $\forall x \in D, P(x) \vee Q(x) \not\equiv (\forall x \in D, P(x)) \vee (\forall x \in D, Q(x))$.

Let $P(x)$: x is even, where $x \in \mathbb{Z}$ and

$Q(x)$: x is odd, where $x \in \mathbb{Z}$.

For any integer x, x is even or x is odd.

i.e. $\forall x \in \mathbb{Z}, P(x) \vee Q(x)$ is true.

However, for any integer x, x is even is false and for any integer x, x is odd is False.

Therefore, $(\forall x \in \mathbb{Z}, P(x)) \vee (\forall x \in \mathbb{Z}, Q(x))$ is false.

$\forall x \in \mathbb{Z}, P(x) \vee Q(x) \not\equiv (\forall x \in \mathbb{Z}, P(x)) \vee (\forall x \in \mathbb{Z}, Q(x))$.

Ex. 01. Determine whether $\forall x \in D, P(x) \Longrightarrow Q(x)$ and $\forall x \in P(x) \Longrightarrow \forall x \in Q(x)$ are logically equivalent. Justify your answer.

02. Show that $\exists x \in D, P(x) \vee Q(x)$ and $(\exists x \in D, P(x)) \vee (\exists x \in D, Q(x))$ are logically equivalent.

03. Determine whether $\exists x \in D, P(x) \wedge Q(x)$ and $(\exists x \in D, P(x)) \wedge (\exists x \in D, Q(x))$ are logically equivalent

- **Negation of conjunction and disjunction with open sentences**

(i) $\neg(\forall x \in D, P(x) \wedge Q(x)) \equiv \exists x \in D, \neg(P(x) \wedge Q(x))$
$\equiv \exists x \in D, (\neg P(x)) \vee (\neg Q(x))$

(ii) $\neg(\forall x \in D, P(x) \vee Q(x)) \equiv \exists x \in D, \neg(P(x) \vee Q(x))$
$\equiv \exists x \in D, (\neg P(x)) \wedge (\neg Q(x))$

(iii) $\neg(\exists x \in D, P(x) \wedge Q(x)) \equiv \forall x \in D, \neg(P(x) \wedge Q(x))$
$\equiv \forall x \in D, (\neg P(x)) \vee (\neg Q(x))$

(iv) $\neg(\exists x \in D, P(x) \vee Q(x)) \equiv \forall x \in D, \neg(P(x) \vee Q(x))$
$\equiv \forall x \in D, (\neg P(x)) \wedge (\neg Q(x))$

Implication and open sentences

If we have two open sentences $P(x)$ and $Q(x)$, and we join them with $\Longrightarrow$, we have seen that the resulting sentence $P(x) \Longrightarrow Q(x)$ is again an open sentence. In order to make a statement of it, we may either plug an element from the domain in for x or use one of the two quantifiers $\forall$or $\exists$.

Statements of the form

$$\forall x \in D, P(x) \Longrightarrow Q(x).$$

are so common place that there are multiple ways to express them.

Here are some common ways this is done.

- Given $x \in D$, if $P(x)$, then $Q(x)$.
- Let $x \in D$. If $P(x)$, then $Q(x)$.
- If x is an arbitrary element of D satisfying $P(x)$, then $Q(x)$.
- If x is an element of D satisfying $P(x)$, then $Q(x)$.

$$\exists x \in D, P(x) \Longrightarrow Q(x).$$

The words we associate with $\exists$ are: “for some", “exists", “at least ", etc.

Eg There exists real number x such that $\underbrace{|x| = x \text{ whenever } x > 0}_{\text{Predicate}}$.

$(\exists x \in \mathbb{R}, \quad x > 0 \Rightarrow |x| = x)$

Negation of the given statement is " For all real number x, $x > 0$ and $|x| \neq x$"

" For every positive real number x, $|x| \neq x$"

Examples:

01. "For an arbitrary integer x , if x is even, then $x - 2$ is even."

Negation of the 01. "There exists an integer x such that x is even and $x-2$ is odd"

Let $P(x)$: x is even and

$Q(x)$: $x - 2$ is even, where $x \in \mathbb{Z}$.

Then, the above statement can be expressed as following way:

$$\text{"}\ \forall x \in \mathbb{Z}, P(x) \Longrightarrow Q(x).\text{"}$$

Similarly the biconditional an integer x is even if and only if $x - 2$ is even

$$\forall x \in \mathbb{Z}, P(x) \Leftrightarrow Q(x).$$

02. The statement "a real number x has a real square root if it is positive" might be interpreted in symbolic logic as

$\forall x \in \mathbb{R}, x$ is positive $\Longrightarrow x$ has a real square root

Or, $\forall x \in \mathbb{R}, x > 0 \Longrightarrow \exists y \in \mathbb{R}, x = y^2$.

Exercise . Translate the following symbolic logic statements into English.

(a) $\exists x \in \mathbb{R},\ x^2 = 2$.

(b) $\forall x \in \mathbb{Z},\ \ x$ is even $\Leftrightarrow\ x^2$ is even .

(c) $\forall x \in \mathbb{R},\ \ x > 1\ \Rightarrow\ x^3 > 1$.

(d) $\forall x \in \mathbb{R},\ \ x^2 - 2x + 1 = 0\ \Rightarrow\ x = 1)$.

(e) $\exists x \in \mathbb{Q},\ 2x^3 - x^2 + 2x - 1 = 0$.

(f) $\forall x \in \mathbb{R}\ \exists y \in \mathbb{Z}\ \exists z \in [0, 1),\ x = y + z$.

For all real number x, there exists an integer y, there exists z in $[0,1)$ such that $x = y+z$.

Note:

01.

Conditional Quantification

- Consider the conditional quantification: $\forall x \in D, P(x) \Longrightarrow Q(x)$
- Then we defined
  - Converse: $\forall x \in D, Q(x) \Longrightarrow P(x)$
  - Contrapositive: $\forall x \in D, \sim Q(x) \Longrightarrow \sim P(x)$
  - Inverse: $\forall x \in D, \sim P(x) \Longrightarrow \sim Q(x)$
- **Note:** A conditional proposition is logically equivalent to its contrapositive.

Negation of Conditional Quantification

$$\begin{aligned} \sim(\forall x \in D, P(x) \Longrightarrow Q(x)) &\equiv \exists x_0 \in D, \sim(P(x_0) \Longrightarrow Q(x_0)) \\ &\equiv \exists x_0 \in D, P(x_0) \wedge \sim Q(x_0) \end{aligned}$$

02. **Statements with multiple quantifiers**

Statements involving multiple quantifiers are quite common in mathematics. For such statements, it is important to understand how the quantifiers interact and to be able to analyze the truth values of the statements. We begin with two examples that demonstrate that the order in which quantifiers occur is important.

Example 01:

Let $P(x, y): x > y$, where x and y are the real numbers.

We examine the quantified statement

$$\forall x \in \mathbb{R}\ \exists\ y \in \mathbb{R}, P(x, y).$$

This statement means that

"For each real number x, there is a real number y such that $x > y$."

Or in other words,

"for each real number x, there is a real number y (possibly depending on x) that is smaller than x."

We claim that this statement is true.

Let x be a real number. Choose $y = x - 1$. Then y is real number and $x > y$.

Therefore, for each real number x, there is a real number y such that $x > y$.

Example 02:

Let $P(x, y): x > y$, where x and y are the real numbers

We next examine the quantified statement

$$\exists y \in \mathbb{R}\ \forall\ x \in \mathbb{R}, P(x, y).$$

i.e. "There is some real number y for any real number x such that, $x > y$."

This statement is false.

Suppose there exists real number y which is greater than for any real number x.

Choose $x = y - 1$. Then x is a real number and x is not greater than y.

The previous two examples demonstrate that changing the order of quantifiers in a statement with multiple quantifiers can change the truth value of the statement. This only happens when the quantifiers are **not the same type**. Changing the order of existential quantifiers that are next to each other, or universal quantifiers that are next to each other, will not change the truth value of the statement.

For example,

$$\forall x \in \mathbb{R}\ \forall\ y \in \mathbb{R}, x > y.$$

has the same truth value as

$$\forall y \in \mathbb{R}\ \forall\ x \in \mathbb{R}, x > y.$$

Since the order does not matter in this case we will abbreviate either of these statements as

$$\forall x, y \in \mathbb{R}, x > y.$$

Which we can read as "for any real numbers x and y, we have $x > y$."

Similarly,

$$\exists x \in \mathbb{R}\, \exists\, y \in \mathbb{R}, x > y.$$

has the same truth value as

$$\exists y \in \mathbb{R}\, \exists\, x \in \mathbb{R}, x > y.$$

Since the order does not matter in this case, we may abbreviate either of these statements as

$$\exists x, y \in \mathbb{R}, x > y.$$

Which we might read as "there are real numbers x and y such that $x > y$."

Multiple quantifiers and negation

Example. Let x, y, and z be variables with domains S, T, and U, respectively, and let $P(x, y, z)$ be an open sentence. We will negate the statement

$$\forall x \in S, \forall y \in T, \exists z \in U,\ P(x, y, z).$$

In order to do this, we will use parentheses to make the order of the quantifiers clearer. Hence, the statement that we wish to negate is

$$\forall x \in S, (\forall y \in T, (\exists z \in U,\ P(x, y, z))).$$

Proceeding one level at a time, we see that

$$\begin{aligned} &\neg(\forall x \in S, \forall y \in T, \exists z \in U,\ P(x, y, z)) \\ \equiv\ & \exists x \in S,\ \neg(\forall y \in T, \exists z \in U,\ P(x, y, z)) \\ \equiv\ & \exists x \in S, \exists y \in T,\ \neg(\exists z \in U,\ P(x, y, z)) \\ \equiv\ & \exists x \in S, \exists y \in T, \forall z \in U,\ \neg P(x, y, z). \end{aligned}$$

Notice that negating this quantified statement was as simple as swapping all existential and universal quantifiers, and negating the open sentence at the end.

EX:

let ε, δ, and x be real variables, and define

$$\lim_{x \to a} f(x) = L$$

$0<|x-a|<\delta$

to mean Given any $\varepsilon>0$ there exists $\delta>0$ for each x in $\mathbb{R}$, $|f(x)-L|<\varepsilon$ whenever

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R},\ 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

What is the negation of the above statement?

$$\neg(\ \forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R},\ 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.)\,?$$

Exercise . Write down the negation of the following statements in symbolic form, and translate them into English.

01.

(a) $\exists x \in \mathbb{R},\ x^2 = 2.$

(b) $\forall x \in \mathbb{Z},\quad x \text{ is even} \ \Leftrightarrow\ x^2 \text{ is even}\ .$

(c) $\forall x \in \mathbb{R},\quad x > 1 \ \Rightarrow\ x^3 > 1\ .$

(d) $\forall x \in \mathbb{R},\quad x^2 - 2x + 1 = 0 \ \Rightarrow\ x = 1).$

(e) $\exists x \in \mathbb{Q},\ 2x^3 - x^2 + 2x - 1 = 0.$

(f) $\forall x \in \mathbb{R}\ \exists y \in \mathbb{Z}\ \exists z \in [0,1),\ x = y + z.$

02. Consider the quantified statement

$$\forall x, y, z \in \mathbb{R},\ (x-3)^2 + (y-2)^2 + (z-7)^2 > 0$$

(a) Express $(*)$ in words.

(b) Is $(*)$ true or false? Explain.

(c) Express the negation of $(*)$ in symbols, and then in words.

(d) Is the negation of $(*)$ true or false? Explain.

End of D6